Erebus 2027 · Match Studio

Match Studio for Webots R2025a · RoboCupJunior Rescue Simulation 2027 draft rules



Quick start · Install & run

Match Studio · 2027 rules · Branch guide · PDF guide

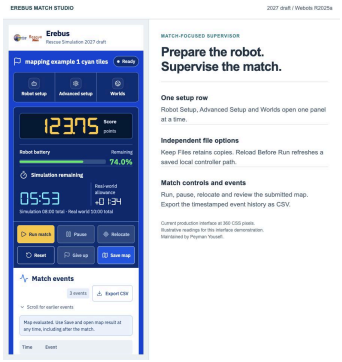
Match Studio brings the **RoboCupJunior Rescue Simulation 2027 draft rules** to the customized Erebus referee interface, with **Webots R2025a** support on Windows, macOS and Linux.

Referee interface Score, clocks, live battery and match controls in one compact window.	**Setup and world selection** Organized Robot Setup, Advanced Setup and searchable field previews.
Interactive 3D replays Playback, timeline seeking, camera controls and a final score breakdown.	**Submitted-map review** Cell-by-cell comparison, error highlighting, PNG export and the original matrix.
Events and HUD Timestamped match history, CSV export and replay-synchronized events.	**Controller file handling** Keep files after Reset and optionally reload local source before each match.
Unauthorized access detection Monitors for unauthorized data transfers between participant code, the supervisor and Webots.	**2027 rules** Motor-based battery consumption, configurable replacement on cyan tiles, smiley victims and updated scoring.

Branch guide

Choose the branch that matches your competition and Webots installation.

Branch	Rules	Webots	Use this for
`2026_webots2023b`	2026	R2023b	Match Studio 26.2: match supervision, replay and map review.
`2027_webots2025a` (this branch)	2027 draft	R2025a	Match Studio with battery consumption, cyan contact tiles and smiley victims.
`2028_webots2025a`	Future development	R2025a	Robot Designer and component library; under development.

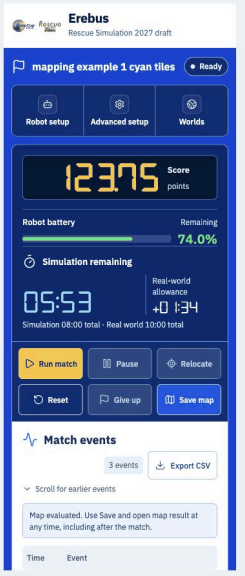


Example views of the supervisor, replay and map-review tools. Setup and battery readings are illustrative; the sample map includes three reporting errors.

Match Studio

Match Studio keeps score, battery, clocks, controls and events together. Setup panels open only when selected, leaving the match view compact.

Feature	What the referee can do
Match console	Read decimal scores, live battery, simulation remaining time and the real-world allowance.
Connected setup panels	Open **Robot Setup**, **Advanced Setup** or **Worlds**; only one panel expands at a time. Search, preview, select and load a bundled field.
Match controls	Run, pause, relocate, reset, give up and save the accepted map.
Interactive HTML replay	Play, pause, seek, rotate and zoom a recorded 3D match in a browser; share its ZIP and final-score summary.
Submitted-map review	Select cells to compare reported and expected values, rotate, zoom, pan, fit, save PNG and export the original matrix.
Match events	Follow timestamped events and export the complete history as chronological CSV.
Webots HUD	Independently show score, time and the latest three events, including in the replay.
File handling	Retain uploaded files after Reset, optionally refresh a local controller source before Run, and recover from missing files or invalid robot JSON.
External controllers	Use the existing remote controls, with visible connection errors and duplicate-start protection.



MATCH-FOCUSED SUPERVISOR

Prepare the robot. Supervise the match.

One setup row

Robot Setup, Advanced Setup and Worlds open one panel at a time.

Independent file options

Keep Files retains copies. Reload Before Run refreshes a saved local controller path.

Match controls and events

Run, pause, relocate and review the submitted map. Export the timestamped event history as CSV.

Current production interface at 360 CSS pixels.
Illustrative readings for this interface demonstration.
Maintained by Peyman Yousefi.

Keep Files and Reload Before Run

These switches are independent:

Keep Files	Reload Before Run	Result
Off	Off	Reset removes loaded copies; load the controller and JSON again.
On	Off	Reuse the retained copies. Editing the original source does not update them.
Off	On	Refresh the controller from its saved path for the next match; JSON still uses normal loading after Reset.
On	On	Retain the copies and refresh the controller for each new local match.

The source path belongs to the **computer running Webots**. Relative paths start in the Erebus project folder. Enter it manually: browser file pickers provide contents, not a portable original full path. Robot JSON keeps its normal browser-loading workflow.

Refresh applies to a **new local Run**, not pause/resume or Remote mode. Supported entries are `.py`, `.exe`, `.class`, `.jar`, `.bsg`, `.m`, or a compiled executable without an extension. C/C++ must already be compiled. Check Robot Setup for missing-file messages and the Webots console for controller errors.

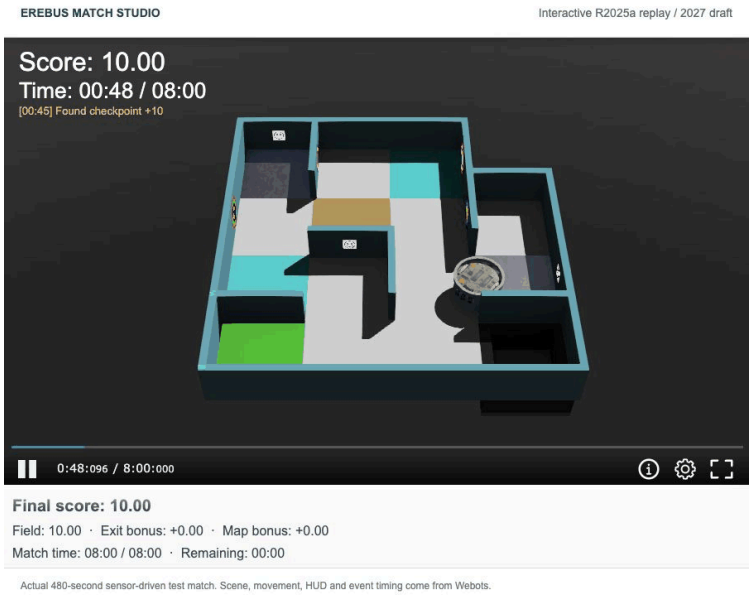
Advanced Setup defaults

Setting	Default
Keep Files	On
HTML replay recording / Webots HUD	On / On
Reload Before Run / Automatic map result	Off / Off
Automatic camera / Keep Remote on Reset	Off / Off
Disable automatic LoP / Enable debugging	Off / Off

Use **Open Compact Window** for a separate, resizable browser window. Allow browser popups if it does not open.

Interactive 3D replay

Recording starts after robot setup and the initial HUD are ready. At completion, timeout, give-up or Reset, Erebus finalizes the recording before removing the robot. Matches are saved as **interactive HTML/W3D replays** with the final score and event history.



The replay includes the 3D scene and animation, final score, field/exit/map breakdown, remaining time and playback-synchronized events. Events follow the playback position when seeking forward or backward.

Open `recording/replay.html` directly in a supported browser, or serve it over HTTP(S). Keep the complete recording folder for archival; the accompanying ZIP contains a `recording/` directory. Scene, animation and exported local assets are embedded, but the Cyberbotics viewer still requires internet access and WebGL.

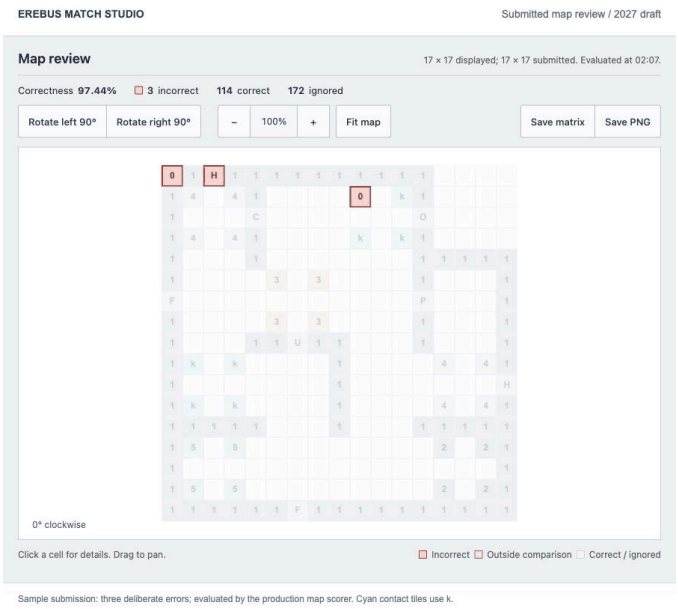
To embed a hosted recording:

```
<iframe src="/matches/example/recording/replay.html"
        title="Erebus match replay" loading="lazy" allowfullscreen
        style="width:100%;height:720px;border:0"></iframe>
```

A hosting website can read `match-result.json` to display score cards. Configure access permissions on the website where you publish recordings.

Submitted-map review

Review the **accepted map submission**, its correctness result and each reported-versus-expected cell. The review is available after the supervisor evaluates a submitted map.



- Correct cells fade; incorrect reports stay prominent. Nonzero content in either map remains visible.
- Select a cell for its original matrix coordinates, reported value, expected value and comparison status.
- Rotate in 90-degree steps, zoom, pan or fit. Save a PNG or export the untouched reported matrix as JSON.
- Only outer margins that are zero in both maps are cropped. Cells outside the scored comparison are marked separately.

Automatic map result saves and opens the review when a map is accepted. With it off, **Save map** still works after evaluation, including after robot removal. Repeated saves use the same match folder; rejected later submissions do not replace the accepted result. Save before Reset.

Events, HUD and saved files

Export CSV downloads all events, oldest first, with `Timestamp` and `Event` columns. UTF-8 and CSV quoting preserve commas, quotation marks and multiline entries. It is available during or after a match when events exist; Reset clears the history. Times are match times, not wall-clock timestamps.

The HUD switch is independent of recording. It controls score, elapsed/maximum time and the latest three events in Webots. Turning visible labels off retains the timing information needed to synchronize replay events.

Each match has its own start-time folder:

```
match-results/<start-time>/
├── recording/
│   ├── replay.html, replay.w3d, replay.json, replay.css, replay.jpg
│   ├── match-result.json
│   ├── textures/                  # When exported
│   └── <start-time>.zip
└── map/
    ├── map-review.html
    ├── map-result.json
    └── reported-map.json
```

Recordings and map reviews are saved independently. The recording ZIP contains the `recording/` folder; archive the `map/` folder separately to keep the map review.

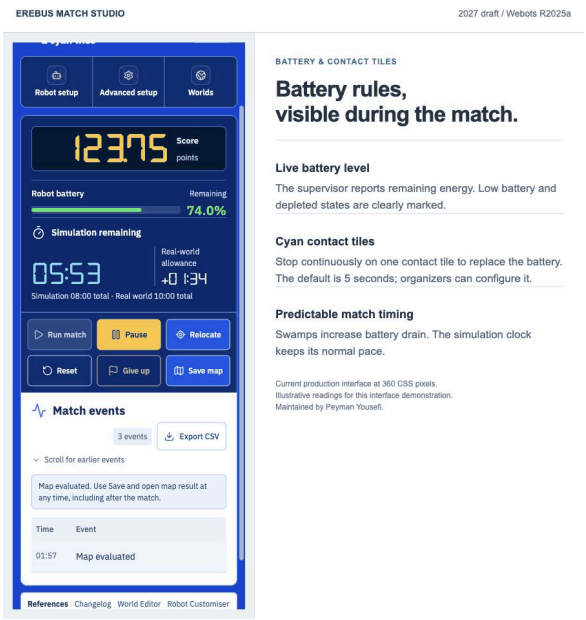
Security and unauthorized access detection

When Wireshark is installed and configured on the referee's computer, Match Studio monitors for unauthorized data transfers between participant code, the Erebus supervisor and Webots, and alerts the referee when detected.

2027 rules

See the RoboCupJunior Rescue Simulation 2027 draft rules for the full competition requirements. Key match settings and reporting requirements are summarized below.

Rule area	Match behavior
Battery consumption	The supervisor measures wheel movement and updates the live battery meter. Additional driven wheels count separately.
Cyan contact tiles	Remain continuously stationary on the **same tile for 5 seconds** to replace the battery. Movement, leaving the tile or relocation resets the interval. The duration is organizer-configurable.
Swamps	Increase battery consumption; **the match clock keeps its normal pace**.
Empty battery	Ends the run. An empty battery cannot be revived by remaining on a contact tile.
Victim appearance	**H = harmed/sad, S = stable/neutral, U = unharmed/happy** smileys. Three-dimensional decoys do not score. Victim signs are 20 mm square; cognitive targets are 50 mm across.
Token reporting	Requires at least **one second stopped**, with the robot and reported position within **60 mm** of the sign. Wrong-category and duplicate reports are handled by the scoring rules.
Mapping	Cyan contact tiles use `k`. Map review uses the accepted submission and the same scorer as the supervisor.
Match duration	Up to **480 simulated seconds**, with a **600-second real-world limit**. Exit, give-up or battery depletion can end a run earlier.
Match ending	Exit eligibility, map bonus and rounding are evaluated by the supervisor. The final map award is included in the completed score.



Example battery display: remaining charge, low battery and replacement on a cyan tile.

Organizer battery configuration

Edit `sensorEnergyConfig.txt` in the MainSupervisor folder and restart the supervisor after changing values:

```
ChangeBatteryTime = 5
Motor_Drain_Per_Step = 0.02,
Motor_Max_Velocity = 6.28,
Battery_Per_Step = 0.00,
SwampMultiplier = 5
```

Set `ChangeBatteryTime` to the required stationary duration in seconds. The default is **5 seconds**.

At the default settings, a two-wheel pair running at 6.28 rad/s consumes 0.02 battery percentage points per reference 16 ms. Additional driven wheels contribute separately. `SwampMultiplier = 5` applies five times the normal consumption in swamps. Passive and sensor consumption default to zero; organizers can adjust these values in the configuration file.

Map scoring

The mapping multiplier is `1 + 1.2 × correctness`, where correctness ranges from 0 to 1. A fully correct map gives a multiplier of **2.2**. Use **Save map** to review the accepted submission and its correctness result.

Competition graphics profile

Set these Webots preferences consistently on competition machines: **Ambient Occlusion Low**, **Texture Quality Low**, **Max Texture Filtering 4**, **Shadows Disabled**, **Anti-aliasing Disabled**.

Battery in the controller interface

A normal `G` **game-info request** receives a 20-byte packet:

```
kind, score, simulation_remaining, real_remaining, battery_percent = \
    struct.unpack('c f i i f', packet)
```

Battery percentage is appended to the original fields. Older readers that require exactly 16 bytes need updating. The supplied `GetGameInfo.py` accepts both packet sizes. Use the standard reporting messages to report victims, submit maps and request game information.

Quick start

What you need

Software	Version	Download / setup
Webots	**R2025a**	Download · Setup
Erebus Match Studio	**2027**	Download ZIP
Python	**3.10 · 64-bit**	Download · Select in Webots
Python packages	requirements.txt	Install below

On Windows, enable **Add Python to PATH** during installation. Select Python 3.10 in **Webots preferences → General → Python command**.

Optional for referees: Wireshark download · Setup. Erebus runs without it; unauthorized-transfer monitoring requires it.

Install Python packages

Extract the Erebus ZIP. Open a terminal in the extracted folder.

Windows

```
py -3.10 -m pip install -r requirements.txt
```

macOS / Linux

```
python3.10 -m pip install -r requirements.txt
```

Packages: NumPy · Requests · termcolor · overrides · Pillow · OpenCV

Start Erebus

1. Open a world file from **game/worlds** in Webots.
2. If Match Studio does not open automatically, right-click **Main Supervisor** in the Webots toolbar and select **Show Robot Window**. Match Studio opens in your default browser.
3. In **Match Studio → Robot Setup**, load your robot program and robot JSON.
4. Press **Run match**.

Example programs · Robot customizer

Robot configuration

Use the existing robot customizer with a **3,000-point component budget** for this edition. Robot Designer and its component library are part of the separate 2028 development branch.

Credits and licensing

- **Original Erebus:** RoboCupJunior Erebus maintainers and contributors, based on `v26.1`.
- **Fork modifications and maintenance:** Peyman Yousefi.
- **2027 reference work:** the official 2027 branch and sebybritez/Bater-a-PRUEBA contributors.
- **Simulation platform:** Cyberbotics Webots.
- **Interface assets:** IBM Plex Sans and DSEG under their bundled SIL Open Font Licenses; Lucide and inherited Feather notices under the bundled ISC notices. Original fonts and Webots models retain their notices.

This is an independent edition of Erebus maintained by Peyman Yousefi. See NOTICE.md and the bundled asset notices for licensing and attribution.

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